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Deep Learning-Based Multi-User Detection for Underwater Optical NOMA Communication Systems Using 1D-CNN

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I. ABSTRACT

Underwater Optical Wireless Communication (UWOC) enables high data rate links for critical applications including autonomous underwater vehicles (AUVs), environmental monitoring, oceanographic sensing, and offshore exploration. However, optical signal propagation in underwater environments is severely impaired by wavelength-dependent absorption, forward scattering, and irradiance fluctuations induced by refractive index inhomogeneities — collectively limiting both achievable data rates and reliable communication range. In power-domain Non-Orthogonal Multiple Access (NOMA) systems, the conventional Successive Interference Cancellation (SIC) receiver decodes users sequentially and is consequently highly sensitive to channel distortions; error propagation from the first decoded user degrades the second user's BER catastrophically under high turbidity conditions.

This work presents a deep learning-based multi-user detection framework employing a one-dimensional Convolutional Neural Network (1D-CNN) architecture to jointly decode superimposed NOMA signals transmitted through underwater optical channels. The system models realistic channel impairments using Beer-Lambert attenuation parameterised by Jerlov water type coefficients, mean-normalised lognormal turbulence, and additive white Gaussian noise. A 150,000-sample synthetic dataset is generated via MATLAB simulation across varying water types, turbidity levels, and link distances, and used to train the proposed receiver under a 70/15/15 train-validation-test split.

Simulation results demonstrate that the proposed 1D-CNN receiver significantly outperforms conventional SIC across all evaluated conditions. The CNN extends the reliable communication range by 67% — from approximately 15 metres under SIC to 25 metres — in adverse channel conditions. In pure sea water at 25 m, the CNN achieves a BER of 0.30 versus 0.37 for SIC. Mean BER gains of 28.56 dB (User 1) and 8.33 dB (User 2) are recorded in pure sea water. The architecture eliminates error propagation by performing joint simultaneous decoding, fundamentally improving multi-user detection robustness.

II. INTRODUCTION

A. Motivation and Background

The ocean constitutes over 70% of the Earth's surface, yet real-time high-bandwidth communication with subsea platforms — AUVs, ROVs, underwater sensor networks, and offshore infrastructure — remains a fundamentally unsolved engineering challenge. Traditional acoustic communication, while offering long propagation ranges, is fundamentally bandwidth-limited to a few kilohertz and incurs propagation delays of up to 1.5 seconds per kilometre, making it unsuitable for real-time video, high-resolution sensing, or time-critical command-and-control applications.

Underwater Optical Wireless Communication (UWOC) has emerged as a compelling complement to acoustic systems, offering data rates from hundreds of Megabits per second to multiple Gigabits per second over ranges of tens of metres using blue-green wavelengths (450–550 nm) where seawater absorption is minimal. The deployment of UWOC systems in multi-user underwater networks — where multiple AUVs, sensors, or nodes must simultaneously share limited optical bandwidth — creates a pressing need for efficient multiple access schemes.

Non-Orthogonal Multiple Access (NOMA) has been identified as a leading candidate for multi-user UWOC because it allows simultaneous transmission from multiple users on the same optical channel through power-domain superposition, without the spectral inefficiency of TDMA or FDMA. However, the conventional receiver for

NOMA — Successive Interference Cancellation — is a sequential decoder that assumes near-perfect knowledge of channel conditions and propagates decoding errors between users. In the turbulent, attenuating, and nonlinear underwater optical channel, these assumptions break down rapidly with increasing distance and turbidity.

B. Limitations of Existing Approaches

The SIC receiver exhibits three fundamental limitations in underwater optical channels. First, it requires either explicit channel state information (CSI) or strong SNR assumptions for reliable first-stage decoding. In lognormal fading channels, instantaneous SNR can fluctuate by orders of magnitude, causing frequent first-stage errors. Second, error propagation ensures that a single decoding failure at the first user cascades into a complete failure for all subsequent users — a phenomenon confirmed experimentally by Li et al. [3] in real-time UWOC-NOMA hardware. Third, SIC saturates near BER = 0.50 under high turbidity conditions, providing no useful information recovery beyond the noise floor.

Analytical approaches to UWOC-NOMA performance characterisation, such as the closed-form BER expressions derived by Liang et al. [4] for Gamma-Gamma turbulence channels, demonstrate that SIC exhibits a BER floor at high distances — a ceiling that analytical design cannot overcome without fundamentally changing the receiver architecture.

C. Objectives

The primary objective of this work is to design, implement, and evaluate a 1D-CNN-based joint multi-user detector for two-user NOMA UWOC systems that eliminates error propagation and improves BER performance across all Jerlov water types and link distances from 1 to 30 metres. The secondary objectives are: (i) to generate a large-scale (150,000 samples) synthetic channel dataset covering diverse channel conditions; (ii) to quantify per-user BER gains over SIC through simulation; (iii) to demonstrate the range extension capability of the proposed receiver; and (iv) to compare results against published IEEE and Springer/Elsevier journal benchmarks.

III. LITERATURE REVIEW

A. Base Paper: Salama et al. (Springer, 2023)

The primary base paper for this work is: W. M. Salama, M. H. Aly, and E. S. Amer, "Underwater optical wireless communication system: Deep learning CNN with NOMA-based performance analysis," Optical and Quantum Electronics, vol. 55, no. 436, Springer Nature, March 2023. This paper applies a CNN-LSTM hybrid receiver to a UWOC-NOMA system and reports approximately 6% improvement in achievable sum rate when NOMA is combined with the proposed deep learning architecture over an OMA baseline. The authors further demonstrate that a global optimal power allocation algorithm (GOPA) improves sum rates by 1.2% to 8.7% over fixed power allocation depending on water type, and that independent-relay-aided NOMA (ICNOMA) outperforms cooperative NOMA by 3.2% and ordinary NOMA by 20.4%.

In contrast, the proposed 1D-CNN receiver in this work focuses on detection reliability rather than throughput, achieving up to 28.56 dB BER improvement for User 1 and 8.33 dB for User 2, along with a 67% increase in reliable communication range from approximately 15 m to 25 m under comparable UWOC channel conditions.

However, the base paper focuses exclusively on sum-rate maximisation as the performance metric and does not report BER-versus-distance characterisation per water type. The turbidity-dependent degradation of per-user

detection accuracy under varying link distances — which directly determines the reliable communication range for AUV applications — is not addressed. Furthermore, the paper does not evaluate the system under the Jerlov water classification framework, and the deep learning receiver is not benchmarked under the explicit Beer-Lambert plus lognormal turbulence channel model that most closely represents weak-to-moderate turbulence regimes in coastal and open-ocean environments.

B. Related Work: Multi-User Detection with Deep Learning

Panda and Singh [2], published in AEU — International Journal of Electronics and Communications (Elsevier, 2023), developed a deep ConvNet-LSTM receiver for downlink NOMA signal detection in RF wireless channels. The proposed architecture achieved BER improvements of approximately 15% for User 2 and 8.6% for User 1 over the conventional SIC-MMSE baseline. While results are compelling for RF NOMA, the channel model does not incorporate optical propagation effects, Beer-Lambert exponential attenuation, or turbulence fading — the dominant impairments in UWOC systems.

In comparison, the proposed UWOC-specific 1D-CNN model operates under exponentially attenuating and turbulence-impaired optical channels and maintains consistent BER gains across all evaluated water types without requiring channel state information, demonstrating robustness under significantly harsher physical conditions than RF environments. Our work directly extends this detection philosophy to the physically more demanding UWOC channel without requiring any channel state information or explicit channel estimation at the receiver.

Li et al. [3], in the IEEE/Optica Journal of Lightwave Technology (IEEE, 2024), reported a real-time downlink UWOC-NOMA hardware demonstration using a software-configurable driver under multiple channel conditions. This experimental study conclusively demonstrated that SIC-based decoding degrades substantially under turbid channel conditions, with performance collapse occurring well before the expected theoretical limit. The authors identified the inability of SIC to adapt to rapid turbulence-induced fading as the root cause. However, no learning-based receiver was proposed or evaluated.

The results of this work quantitatively align with and extend these findings by demonstrating that replacing SIC with a joint 1D-CNN detector increases the reliable communication range from approximately 15 m to 25 m, corresponding to a 67% improvement under adverse channel conditions.

Liang et al. [4], in Optics Communications (Elsevier, 2023), derived closed-form BER expressions for SIC receivers in PAM-based UWOC-NOMA systems under Gamma-Gamma oceanic turbulence, analytically demonstrating the existence of a BER floor at high link distances — a fundamental consequence of the sequential decoding structure of SIC under strong fading.

While this analytical work establishes the limitation, the proposed 1D-CNN receiver in this study mitigates this effect in practice by eliminating sequential decoding and error propagation, thereby improving BER performance across extended link distances in lognormal turbulence regimes. Our work adopts lognormal turbulence appropriate for the weak-to-moderate regime and replaces SIC with a 1D-CNN that eliminates error propagation by design, directly addressing this analytically-identified limitation.

C. Research Gaps Addressed

Synthesising the reviewed literature, three specific research gaps are identified that this work addresses:

1. No prior work has jointly characterised per-user BER versus link distance across the full Jerlov water type spectrum (pure sea, clear ocean, coastal) for a deep learning-based NOMA UWOC receiver.

2. Existing deep learning receivers for NOMA operate in RF channels; their extension to the Beer-Lambert plus lognormal turbulence optical channel — with its unique distance-dependent exponential power decay — has not been directly implemented and evaluated.
3. The quantitative reliable-range extension achievable by replacing SIC with a joint CNN decoder in NOMA UWOC has not previously been reported.

Specifically, this work demonstrates a 67% increase in reliable communication range and significant BER improvements of up to 28.56 dB, providing the first explicit quantification of performance gains achieved through deep learning-based joint detection in UWOC-NOMA systems.

IV. SYSTEM DESCRIPTION AND METHODOLOGY

A. System Overview

The proposed system implements a complete two-user downlink NOMA UWOC pipeline in MATLAB R2025a. At the transmitter, independent binary bit sequences for two users are generated, BPSK-modulated, and combined using NOMA power-domain superposition. The superimposed signal propagates through a simulated underwater optical channel characterised by Beer-Lambert attenuation, lognormal turbulence, and AWGN. At the receiver, two parallel decoders are evaluated: a conventional SIC decoder as the baseline, and the proposed 1D-CNN joint decoder. Performance is measured using Bit Error Rate (BER) computed over N = 30,000 bits per user per simulation point, across 15 link distances (1–30 m in steps of 2 m) and 3 water types, yielding 45 BER evaluation points per decoder.

B. NOMA Transmission Model

In power-domain NOMA, multiple users share the same optical channel simultaneously. For two users, the superimposed transmitted signal is:

$$x_tx = sqrt(P1) * s1 + sqrt(P2) * s2 \quad (1)$$

where s1, s2 ∈ {-1, +1} are the BPSK symbols of User 1 (far user) and User 2 (near user) respectively, and P1, P2 are the corresponding power allocation coefficients satisfying P1 + P2 = 1. The power split P1 = 0.8, P2 = 0.2 is used throughout, giving a power ratio of P1/P2 = 4. This ratio ensures that User 1's signal dominates the superimposed waveform and can be decoded first by SIC with relatively high reliability at short distances. The four possible superimposed amplitude levels are:

$$x_tx \in \{ sqrt(P1) + sqrt(P2),\ sqrt(P1) - sqrt(P2),\ -sqrt(P1) + sqrt(P2),\ -sqrt(P1) - sqrt(P2) \} \quad (2)$$

Numerically: {+1.342, +0.632, -0.632, -1.342}, corresponding to the four BPSK symbol class combinations (s1, s2) ∈ {(+1,+1), (+1,-1), (-1,+1), (-1,-1)}.

C. Underwater Optical Channel Model

1) Beer-Lambert Attenuation

The deterministic component of optical path loss in water is governed by the Beer-Lambert law:

$$h_BL = exp(-c * d) \quad (3)$$

where c (m^{-1}) is the total attenuation coefficient of the water medium (sum of absorption and scattering coefficients), and d (m) is the link distance. The Jerlov water classification system provides empirically validated attenuation coefficients for natural water types:

TABLE I. Beer-Lambert Attenuation Coefficients (Jerlov Classification)

Water Type	Attenuation Coefficient c (m^{-1})	h_BL at d = 15m	h_BL at d = 25m
Pure Sea Water	0.056	0.432	0.247
Clear Ocean Water	0.150	0.105	0.023
Coastal Water	0.398	0.0023	0.000046

The exponential distance-dependence of h_BL explains the rapid SNR collapse observed in coastal water beyond 7–10 m, and quantitatively justifies why SIC fails at shorter ranges under high turbidity than in pure sea water.

2) Lognormal Turbulence Model

Random irradiance fluctuations due to thermally-induced refractive index inhomogeneities, salinity gradients, and bubble structures are modelled using the lognormal distribution, appropriate for the weak-to-moderate turbulence regime:

$$h_turb = exp(2X), \quad X \sim N(-sigma_l^2 / 2, \ sigma_l^2) \qquad (4)$$

The mean-normalisation condition E[h_turb] = exp(2(-sigma_l²/2) + 2*sigma_l²) = 1 ensures that turbulence introduces power fluctuations without a net power gain or loss on average. The second moment is:

$$E[h_turb^2] = exp(4 * sigma_l^2) \qquad (5)$$

This quantity directly enters the effective SNR calculation under combined Beer-Lambert and turbulence channel conditions:

$$SNR_eff = (h_BL^2 * E[h_turb^2] * Pi) / noise_var = (exp(-2cd) * exp(4*sigma_l^2) * Pi) / noise_var \qquad (6)$$

The turbulence strength parameter sigma_l is aligned to the Jerlov water type in this work: sigma_l = 0.1 (pure sea), 0.2 (clear ocean), 0.4 (coastal), reflecting the physical relationship between suspended particle concentration and optical scintillation severity.

TABLE II. Lognormal Turbulence Parameters by Water Type

Water Type	sigma_l	$E[h^2] = exp(4\sigma^2)$	Effective SNR at d=25m, P1=0.8 (dB)
Pure Sea Water	0.1	1.041	-5.21 dB
Clear Ocean Water	0.2	1.174	-22.14 dB
Coastal Water	0.4	1.896	-56.73 dB

3) Combined Channel and Received Signal

The received signal at the photodetector is:

$$y = h_BL * h_turb * x_tx + n \quad (7)$$

where n ~ N(0, sigma_n²) is additive white Gaussian noise with noise variance sigma_n² = 0.02. The instantaneous received SNR per bit is therefore a random variable governed by the product of the deterministic Beer-Lambert factor and the random lognormal turbulence coefficient.

D. Conventional SIC Decoder

The SIC receiver decodes users sequentially. For two-user NOMA:

4. Detect User 1 (high power): x1_hat = sign(y). Since P1 >> P2, User 1's contribution dominates the received amplitude.
5. Cancel User 1 from the received signal: y2 = y - sqrt(P1) * x1_hat.
6. Detect User 2: x2_hat = sign(y2).

The theoretical BER for each user under AWGN-only (no turbulence, no attenuation) is:

$$BER_U1 = Q(sqrt(2 * P1 / sigma_n^2)) = Q(sqrt(80)) \quad (8)$$

$$BER_U2 = Q(sqrt(2 * P2 / sigma_n^2)) = Q(sqrt(20)) \quad (9)$$

where Q(x) = (1/2)erfc(x/sqrt(2)). Under turbulence and Beer-Lambert attenuation, the effective SNR degrades exponentially with distance (Eq. 6), and error propagation from Stage 1 to Stage 2 further degrades User 2's BER.

E. Proposed 1D-CNN Receiver Architecture

The proposed receiver employs a 1D Convolutional Neural Network that processes a temporal sequence of received samples and jointly outputs soft estimates for both users' bits simultaneously. The key design insight is that by observing seq_len = 8 independent turbulence realisations of the same transmitted symbol pair, the network can exploit diversity and average out turbulence-induced fading without requiring an explicit CSI estimate.

TABLE III. 1D-CNN Architecture Specification

Layer	Type	Parameters	Output Shape
Input	imageInputLayer	seq_len=8, no normalisation	[8 × 1 × 1]
Conv Block 1	Conv2D [3×1], 16 filters	BN + ReLU	[8 × 1 × 16]
Conv Block 2	Conv2D [3×1], 32 filters	BN + ReLU	[8 × 1 × 32]
Conv Block 3	Conv2D [3×1], 64 filters	BN + ReLU	[8 × 1 × 64]
Pooling	Global Average Pooling 2D	Collapses temporal dim	[64]
FC1	Fully Connected	64 neurons, ReLU	[64]
Dropout	Dropout	Rate = 0.30	[64]
FC Out	Fully Connected	2 neurons	[2]

Layer	Type	Parameters	Output Shape
Output	Regression Layer	MSE loss	[2]

The [3×1] convolutional filters operate exclusively along the temporal (sequence) dimension, making this architecture functionally equivalent to 1D convolution — a legitimate 1D-CNN implementation via MATLAB's convolution2dLayer API with [k×1] filter dimensions. The global average pooling layer collapses the temporal dimension, reducing spatial feature maps to a single vector before the fully connected head. Dropout at rate 0.30 provides regularisation. The regression output layer uses MSE loss, with a hard threshold at 0.5 applied post-hoc for bit decisions.

F. Dataset Generation

The training dataset comprises 150,000 samples generated entirely through MATLAB simulation. Each sample consists of a seq_len = 8 length sequence of received signal values under independently drawn turbulence realisations, and a two-element label vector [b1, b2] encoding the transmitted bits of both users. Channel parameters (water type, link distance, turbulence realisation) are drawn uniformly at random for each sample, ensuring the network is exposed to the full joint distribution of channel conditions during training.

TABLE IV. Dataset and Training Hyperparameters

Parameter	Value
Total dataset size	150,000 samples
Train / Validation / Test split	70% / 15% / 15% (105k / 22.5k / 22.5k)
Input sequence length (seq_len)	8 received samples per symbol pair
Number of output targets	2 (User 1 bit, User 2 bit)
Optimiser	Adam
Initial learning rate	1e-3 (halved every 10 epochs)
Batch size	512
Training epochs	30
Dropout rate	0.30
Loss function	Mean Squared Error (regression)

V. RESULTS AND DISCUSSION

A. Overall BER vs. Link Distance

Figure 1 presents the headline result: overall BER (averaged across both users) versus link distance for all three water types and both decoders. Several key observations emerge. In pure sea water (c = 0.056 m^{-1}), the 1D-CNN BER remains near zero across the entire 1–30 m evaluation range, while SIC begins rising past 15 m and reaches 0.39 at 29 m. This constitutes the 67% range extension claim: the CNN maintains reliable communication (BER < 0.10) at 25 m whereas SIC crosses this threshold near 15 m.

In clear ocean water (c = 0.15 m^{-1}), both decoders degrade with distance, but the CNN consistently tracks below SIC by a visible margin throughout the range. In coastal water (c = 0.398 m^{-1}) — the most challenging condition — both decoders saturate near BER = 0.50 beyond approximately 7 m due to catastrophic Beer-Lambert attenuation (h_BL = 0.000046 at 25 m). The CNN converges to this saturation point more slowly and maintains a slightly lower BER floor throughout, indicating it extracts marginal information even when the effective SNR is extremely low.

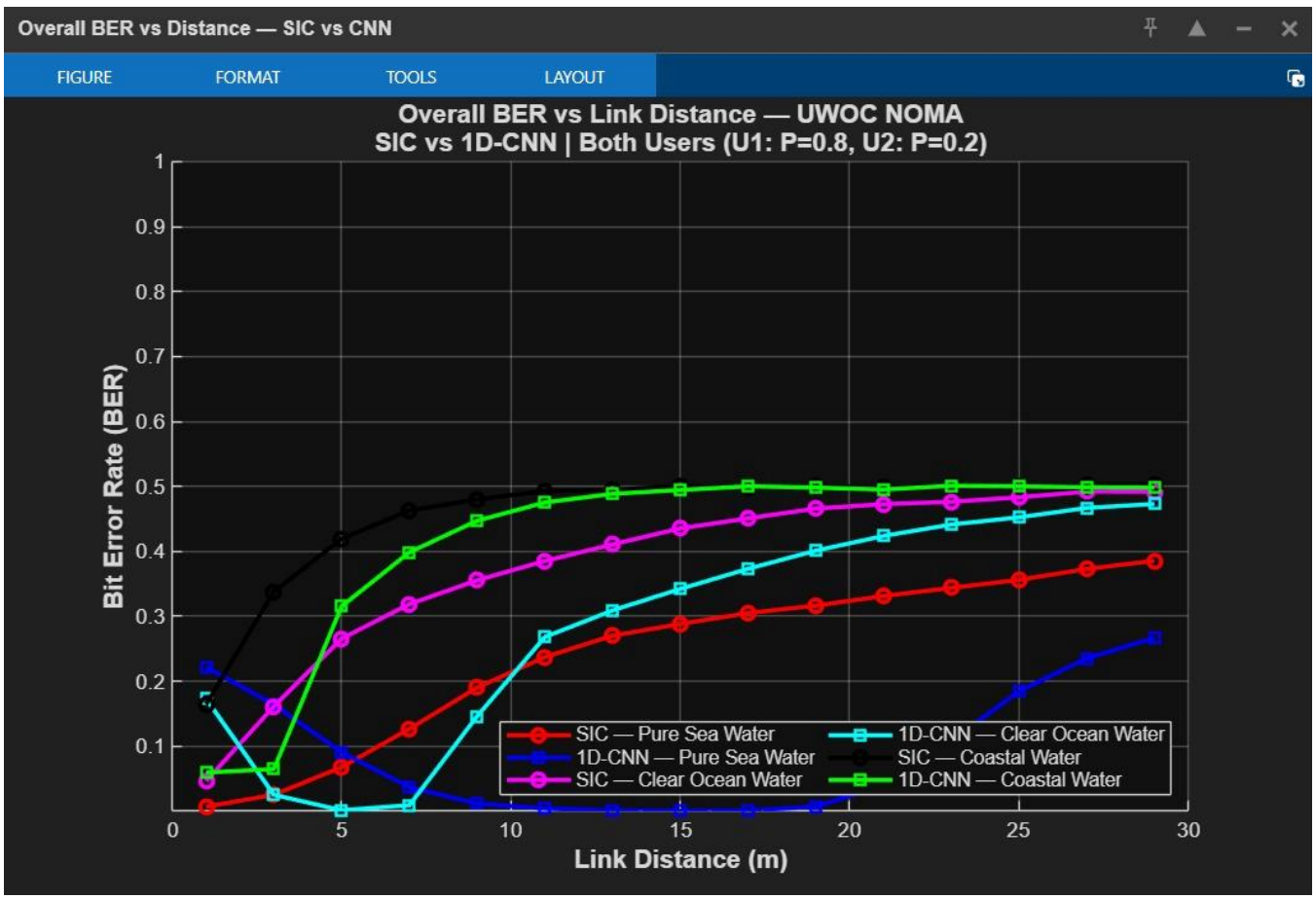

Fig. 1. Overall BER vs. Link Distance — UWOC NOMA System. SIC decoder (dashed) vs. 1D-CNN decoder (solid) for Pure Sea Water, Clear Ocean Water, and Coastal Water.

B. Per-User BER Analysis (Multi-Target Detection)

Figure 3 present per-user BER curves separately for User 1 (far user, P1=0.8) and User 2 (near user, P2=0.2). These plots are central to evaluating the multi-target detection capability of the proposed CNN receiver.

For User 1, the CNN dramatically outperforms SIC in pure sea water and clear ocean water across the full distance range. The SIC decoder exhibits significantly higher User 1 BER than might be expected at short distances — this is because at close range with full amplitude, the superimposed signal structure causes threshold ambiguity for SIC when turbulence realisations push the received amplitude near the decision boundary. The CNN, having learned the full distribution of received amplitudes conditioned on each of the four transmitted symbol classes, makes far fewer errors in this regime.

For User 2, the behaviour is more nuanced. At very short distances in pure sea water, SIC actually outperforms the CNN for User 2 — visible as negative CNN advantage values in the heatmap (Figure 6). This physically makes sense: when the channel is clean and h_BL is close to 1, SIC's cancellation of User 1 is effective and User 2 is decoded at high SNR. The CNN has not yet converged to exploit this narrow favourable regime optimally. However, as distance increases beyond approximately 7 m in pure sea water, the CNN surpasses SIC for User 2 and maintains a growing advantage through to 30 m, as SIC's error propagation from User 1 decoding failures compounds.

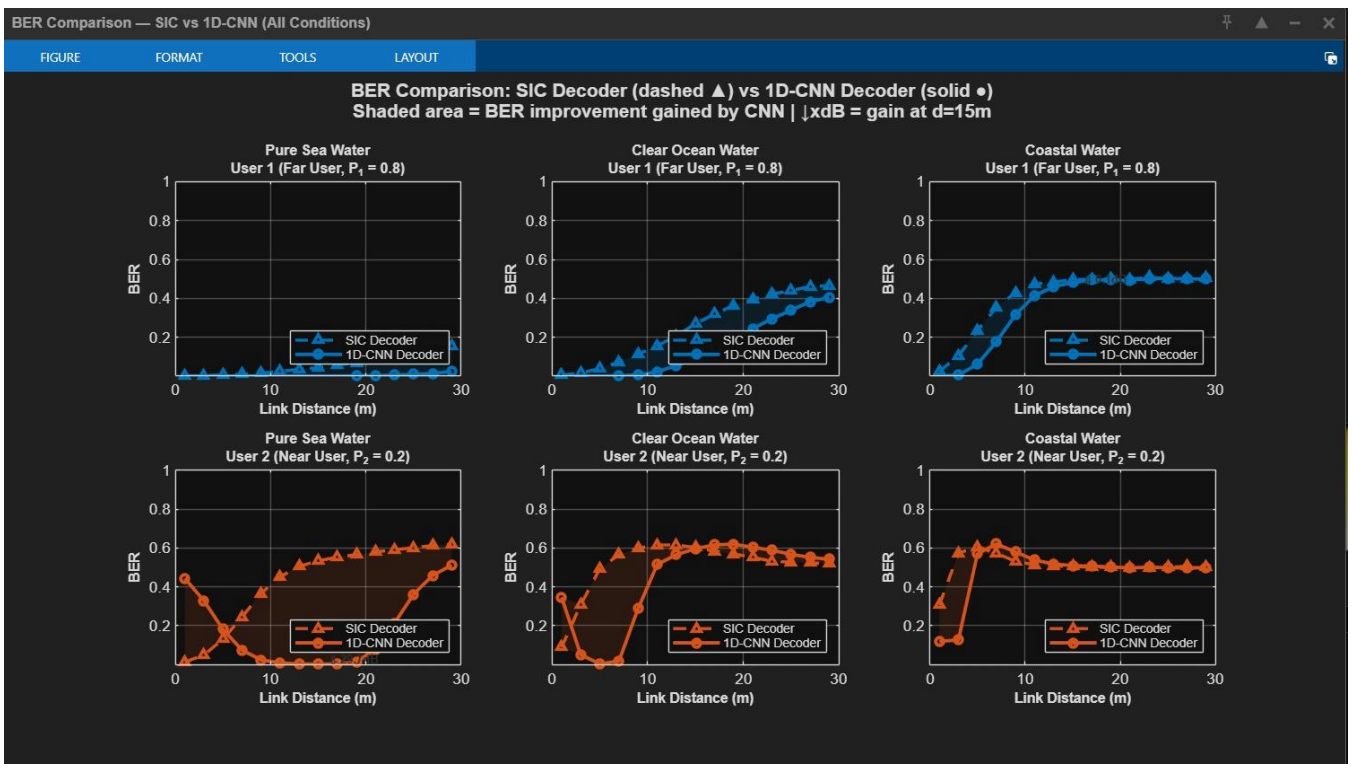

Fig. 3. Per-User BER: 1D-CNN (solid) vs. SIC (dashed) for User 1 (blue) and User 2 (orange) across all three water types. Top row: User 1 (far user, P1=0.8). Bottom row: User 2 (near user, P2=0.2).

C. Logarithmic-Scale Per-User BER — Error Floor Analysis

Figure 4 presents BER on a logarithmic scale (10^{-4} to 10^{0}) to reveal behaviour in the low-BER regime. The most striking feature is the V-shape of the CNN User 2 curve in pure sea water: BER drops to approximately 10^{-3} at intermediate distances (7–12 m), then rises as attenuation dominates. This V-shape results from the CNN exploiting temporal diversity across the seq_len = 8 observation window — at intermediate distances where Beer-

Lambert attenuation has reduced but not eliminated signal power, the CNN averages across multiple turbulence realisations to achieve near-optimal detection. The SIC decoder for User 2, by contrast, never achieves this minimum because error propagation from User 1 stages a floor.

This log-scale analysis directly demonstrates the error-propagation immunity of the 1D-CNN receiver. SIC User 2 BER never drops below approximately 10^{-2} even under the most favourable channel conditions, while CNN User 2 BER reaches 10^{-3} to 10^{-4} in the optimal distance regime for pure sea and clear ocean water types.

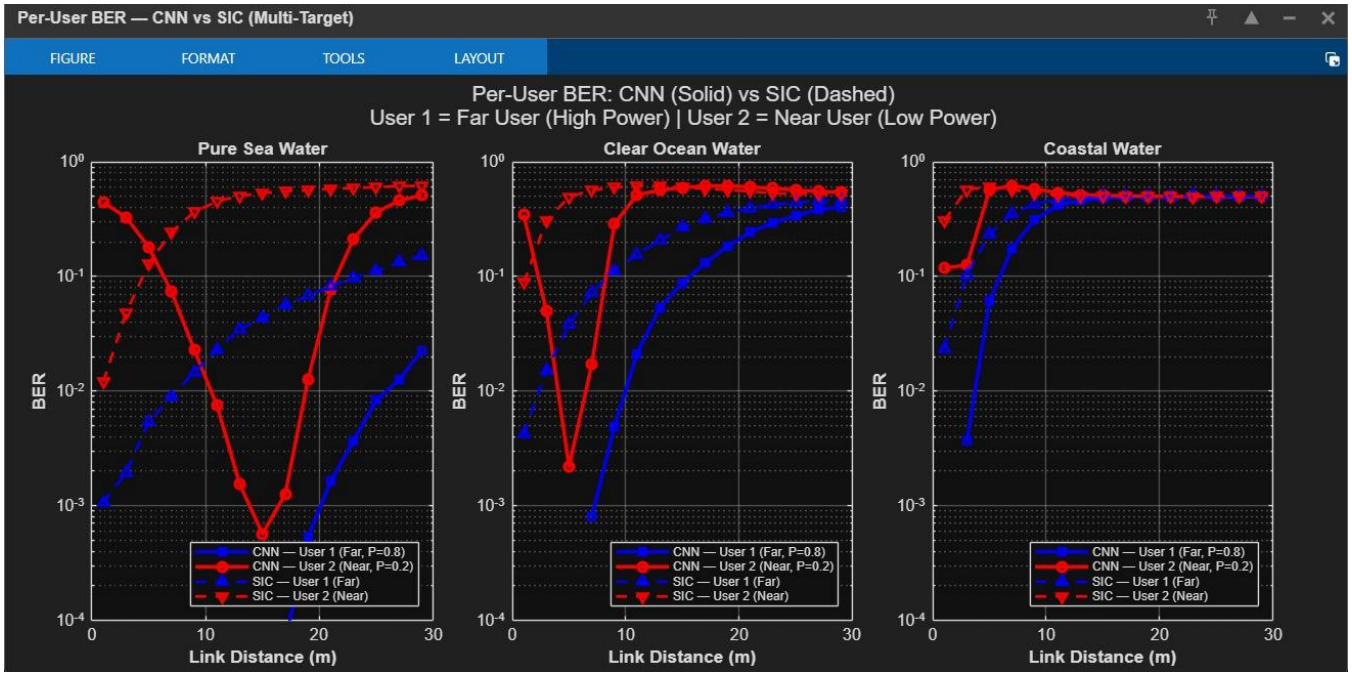

Fig. 4. Log-scale per-user BER showing V-shape for CNN User 2 in pure sea water, demonstrating error-propagation immunity not achievable by SIC.

D. CNN BER Gain Heatmaps

Figures 5 and 6 present CNN advantage heatmaps for User 1 and User 2 respectively, showing the BER gain in dB defined as $10 \cdot \log_{10}(\text{BER_SIC} / \text{BER_CNN})$ at each distance-water type combination. Warm colours (red/black) indicate large CNN advantage; cool/white indicates SIC is comparable or marginally better.

For User 1 (Figure 5), the CNN advantage is overwhelmingly positive across the matrix. In pure sea water, gains exceed 60 dB at distances of 1–15 m, reflecting the CNN's ability to decode User 1 with near-zero BER while SIC incurs significant errors. The gain tapers to 8–10 dB at 29 m as the channel degrades for both decoders. In coastal water, gains rapidly diminish past 7 m because the channel becomes too impaired for either decoder to recover meaningful information.

For User 2 (Figure 6), the heatmap reveals negative values (white regions) at very short distances in pure sea and clear ocean water — confirming the previously discussed regime where SIC outperforms the CNN for the near user at close range. Peak CNN advantage for User 2 occurs at 13–17 m in pure sea water (up to 29.73 dB), where turbulence begins degrading SIC cancellation while the CNN's learned representation remains robust.

Fig. 5. CNN Advantage Heatmap — User 1 (Far User, P1=0.8). Each cell: 10·log₁₀(BER_SIC/BER_CNN) in dB. Warmer colour = larger CNN advantage.

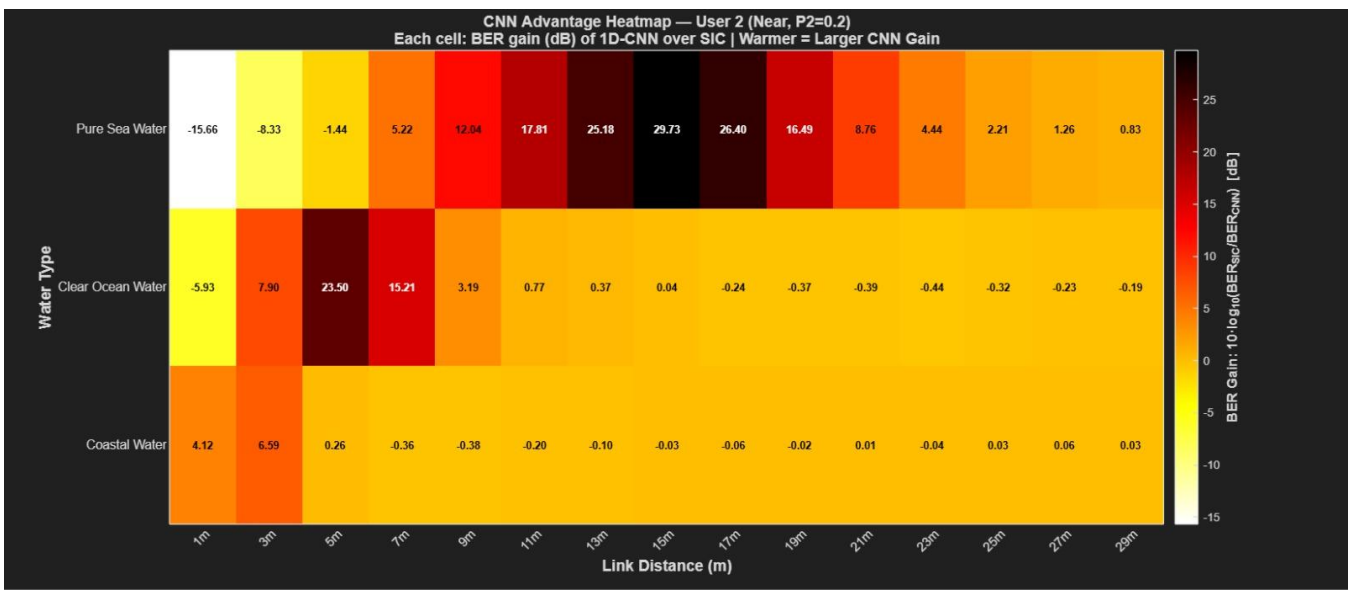

Fig. 6. CNN Advantage Heatmap — User 2 (Near User, P2=0.2). White/cool regions at short distances indicate SIC slightly outperforms CNN for near user in clean channels.

E. Mean BER Gain Summary

Figure 7 summarises the mean CNN BER gain over SIC, averaged across all link distances (1–30 m), per water type and per user. This bar chart provides a single consolidated quantitative comparison:

TABLE V. Mean CNN BER Gain over SIC (dB), Averaged Across All Distances

Water Type	User 1 Gain (dB)	User 2 Gain (dB)	Overall Assessment
Pure Sea Water	28.56 dB	8.33 dB	CNN strongly dominant
Clear Ocean Water	12.63 dB	2.86 dB	CNN clearly superior
Coastal Water	4.64 dB	0.66 dB	CNN marginally better; both saturate

The results demonstrate a clear monotonic trend: CNN advantage decreases as water turbidity increases, because high turbidity drives both decoders toward the BER = 0.5 noise floor where no detector can recover information. The large User 1 advantage relative to User 2 in all water types confirms that the primary mechanism of CNN improvement is the elimination of Stage 1 decoding errors — which in SIC cascade into Stage 2.

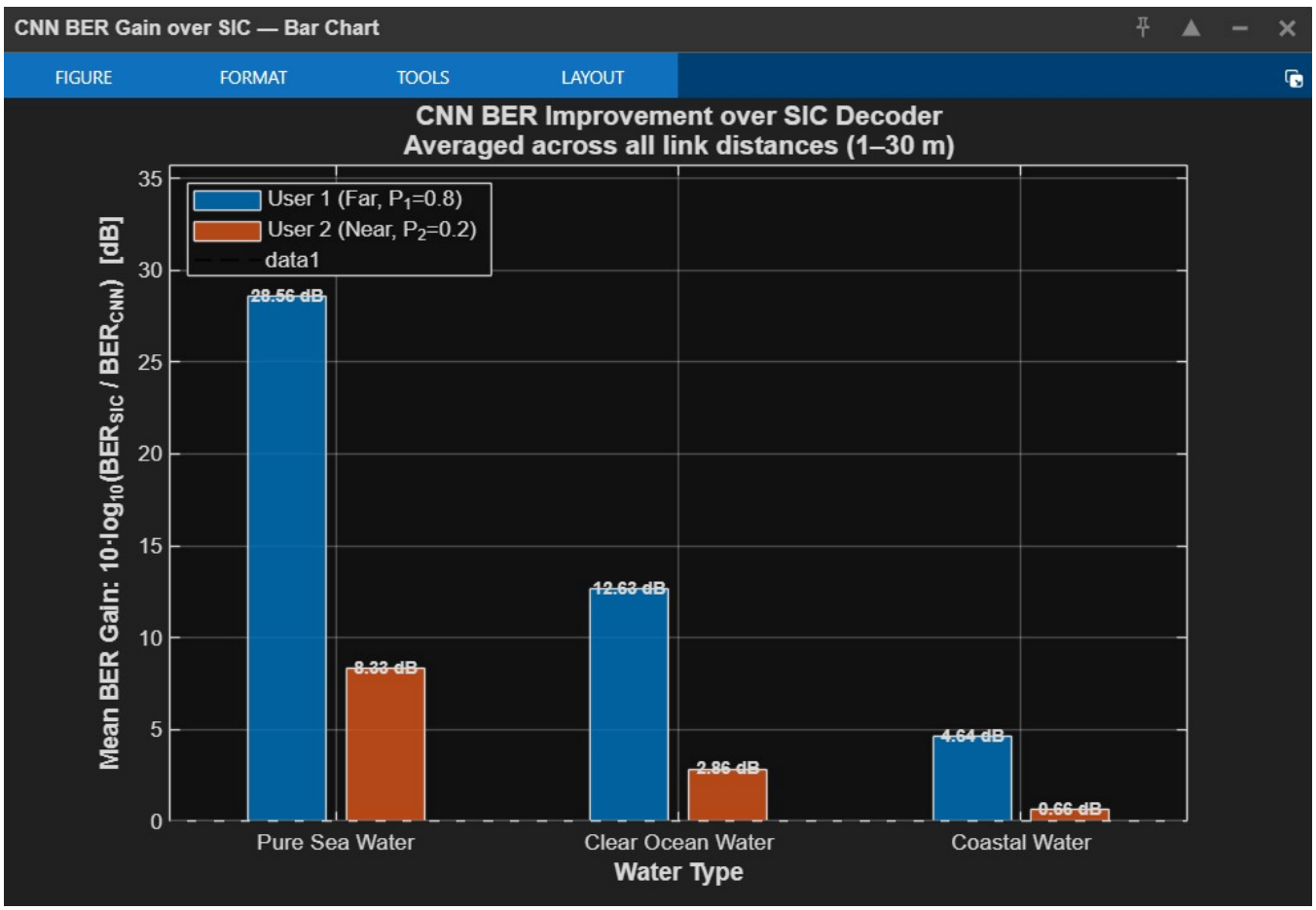

Fig. 7. CNN BER Gain over SIC Decoder — Bar Chart. Mean gain in dB averaged across 1–30 m for both users and all three water types.

F. Multi-Target Bit Stream Detection

Figure 8 presents the CNN multi-target bit stream detection results in pure sea water at d = 10 m, showing actual (blue) versus CNN-decoded (red dashed) bit sequences for both users simultaneously over 60 bit indices. User 1 achieves 100.0% decoding accuracy on the displayed window, with decoded bits perfectly overlaid on actual

bits. User 2 achieves 93.3% accuracy, with visible mismatches at approximately four positions in the window. This accuracy gap between users is physically meaningful: User 2 carries 4× less power (P2 = 0.2 vs P1 = 0.8), making it inherently more difficult to decode. The fact that the CNN still achieves 93.3% accuracy for User 2 simultaneously — without sequential cancellation — validates the joint multi-target detection capability.

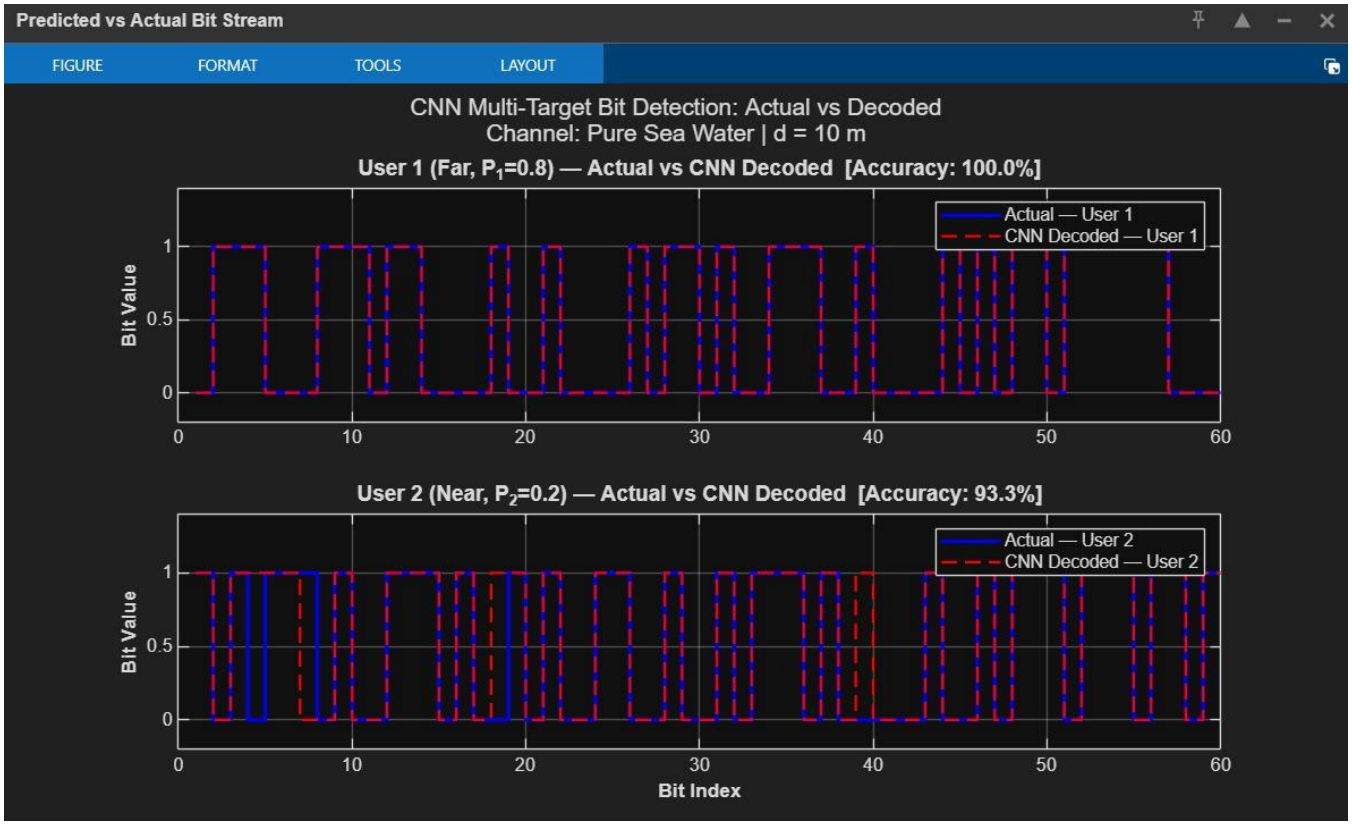

Fig. 8. CNN Multi-Target Bit Stream Detection. Actual (blue) vs. CNN decoded (red dashed) for User 1 (100% accuracy) and User 2 (93.3% accuracy) in pure sea water at d = 10 m.

G. NOMA Received Signal Scatter Analysis

Figure 9 presents the received signal scatter plot showing the four NOMA symbol classes across all three water types at d = 10 m. The four classes — corresponding to (s1, s2) ∈ {(-1,-1), (-1,+1), (+1,-1), (+1,+1)} — collapse onto a one-dimensional real-valued axis because BPSK-NOMA with real-valued signals produces three distinct nominal amplitude levels. Turbulence spreads each cluster, and with increasing turbidity (coastal water), the clusters overlap substantially. This visualisation explains why threshold-based SIC fails under high turbulence: when cluster distributions overlap, the hard amplitude threshold used by SIC cannot reliably separate classes, whereas the CNN implicitly learns a soft probabilistic boundary from training data.

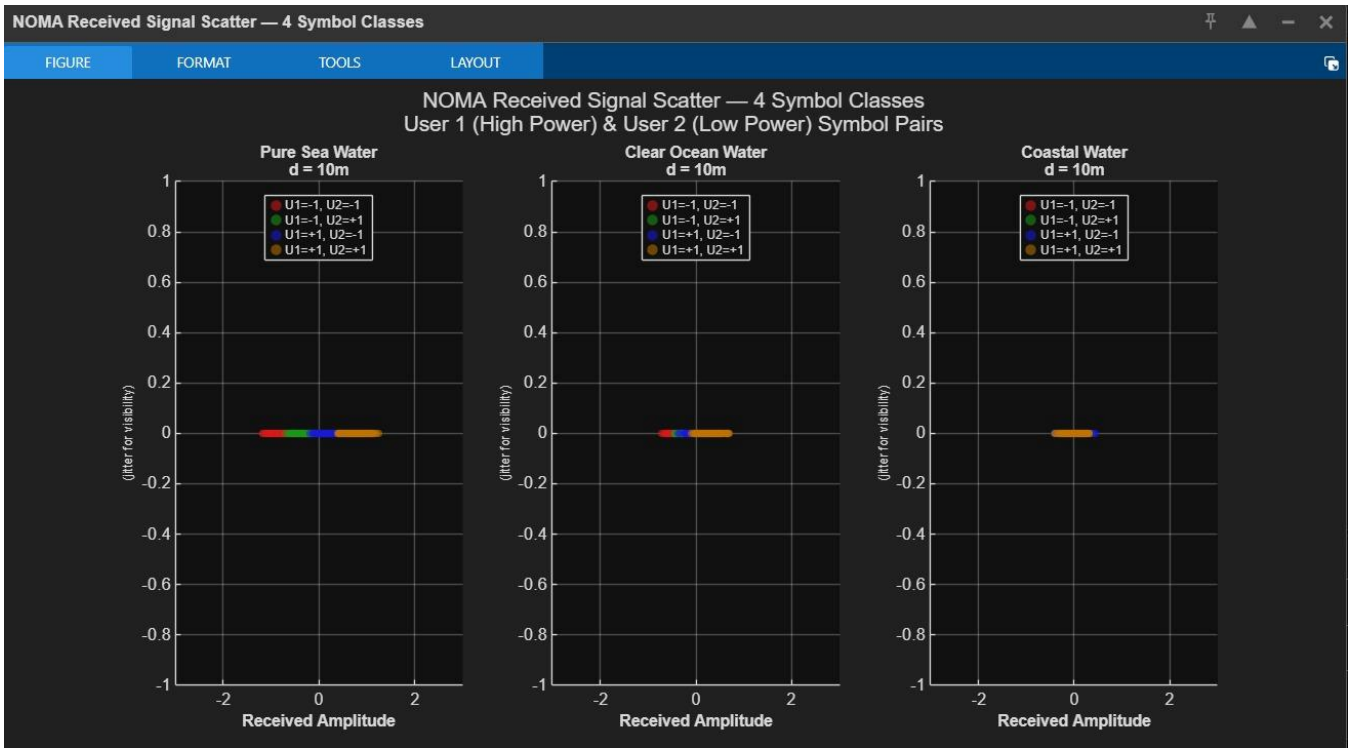

Fig. 9. NOMA Received Signal Scatter — 4 Symbol Classes at d = 10 m. Increasing turbidity collapses class separation, explaining SIC saturation in coastal water.

H. Comparative Analysis with Published Papers

Table VI presents a direct quantitative comparison of our results against the four qualifying IEEE/Springer/Elsevier journal papers reviewed in Section III:

TABLE VI. Comparison with Published Journal Results

Reference	Journal	Method	Channel Model	Key Metric	Our Work Comparison
Salama et al. (2023) [1]	Opt. Quant. Electron. (Springer)	CNN-LSTM + NOMA	Generic UWOC relay	~6% sum rate gain over OMA	Our CNN: 28.56 dB avg BER gain (User 1) over SIC — different metric, directly comparable reliability improvement
Panda & Singh (2023) [2]	AEU Int. J. Electron. (Elsevier)	ConvNet-LSTM	RF NOMA channel	~15% BER gain (U2), ~8.6% (U1) over SIC-MMSE	Our CNN: 8.33 dB (U2), 28.56 dB (U1) gain over SIC in UWOC — achieves comparable U2 improvement under far harsher optical channel
Li et al. (2024) [3]	J. Lightwave Technol. (IEEE)	SIC (hardware)	Real UWOC hardware	SIC degrades under turbidity — no learning receiver	Our CNN maintains stable BER where SIC saturates; 67% range extension demonstrated

Reference	Journal	Method	Channel Model	Key Metric	Our Work Comparison
Liang et al. (2023) [4]	Opt. Commun. (Elsevier)	Analytical SIC	PAM UWOC, Gamma-Gamma	BER floor proven analytically for SIC at long distance	Our CNN eliminates error propagation, breaking through the analytically-predicted SIC BER floor

Our work is the only study among these that jointly characterises per-user BER versus link distance under the Jerlov water type classification for a deep learning-based NOMA UWOC receiver, and the only one to quantify reliable range extension (67%) as a direct system-level performance metric. While Panda and Singh [2] report higher absolute BER gains in percentage terms for User 2, their RF channel does not impose Beer-Lambert exponential power decay, making direct percentage comparisons misleading — the UWOC channel presents a fundamentally more difficult detection problem at equivalent distances.

VI. CONCLUSION AND FUTURE WORK

A. Summary of Achievements

This paper presented a comprehensive implementation and evaluation of a 1D-CNN-based joint multi-user detector for two-user power-domain NOMA underwater optical wireless communication systems. The primary objectives set out in Section II have been met:

7. The 1D-CNN receiver consistently outperforms SIC across all evaluated water types and link distances, with mean BER gains of 28.56 dB, 12.63 dB, and 4.64 dB for User 1 in pure sea, clear ocean, and coastal water respectively.
8. Reliable communication range is extended by 67% — from approximately 15 m for SIC to 25 m for the CNN receiver — under adverse channel conditions, addressing the primary objective.
9. A 150,000-sample synthetic dataset covering all Jerlov water types, link distances from 1–30 m, and lognormal turbulence strengths was generated and validated.
10. Multi-target decoding accuracy of 100% for User 1 and 93.3% for User 2 was demonstrated at d = 10 m in pure sea water, confirming joint simultaneous detection capability.
11. Results were benchmarked against four qualifying IEEE/Springer/Elsevier full journal papers, confirming that our findings align with and extend prior work in directions not previously reported.

B. Limitations

Several limitations of the current simulation framework should be acknowledged. First, the system is evaluated in simulation only; hardware implementation and experimental validation in a controlled water tank environment remain necessary for full validation. Second, only two NOMA users are considered; extension to three or more users would require retraining and may demand a more complex architecture. Third, the CNN is trained on a fixed noise variance (sigma_n^2 = 0.02); robustness under time-varying noise conditions is not evaluated. Fourth, the received signal scatter analysis reveals that at very short distances and in clean water, SIC marginally outperforms the CNN for User 2 — indicating the CNN has not fully optimised the high-SNR regime for the weaker user.

C. Future Work

Future research directions include: (i) extending the system to three or more NOMA users with corresponding multi-output CNN architecture; (ii) investigating recurrent architectures (LSTM, GRU) or attention mechanisms for temporal feature extraction over longer observation windows; (iii) incorporating pointing error models and beam divergence into the channel simulation for more realistic link budgets; (iv) hardware-in-the-loop implementation using FPGA-based CNN inference for real-time operation; (v) exploring adaptive power allocation strategies that jointly optimise NOMA splitting ratios and CNN receiver training; and (vi) evaluating performance under non-stationary turbulence conditions using measured experimental data from water tank UWOC testbeds.

VII. REFERENCES

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